arXiv:2507.07935v6 [cs.AI] 22 Dec 2025

Working with AI: Measuring the Applicability of Generative AI to Occupations*

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Abstract

With generative AI emerging as a general-purpose technology, understanding its economic effects is among society's most pressing questions. Existing studies of AI impact have largely relied on predictions of AI capabilities or focused narrowly on individual firms. Drawing instead on real-world AI usage, we analyze a dataset of 200k anonymized conversations with Microsoft Bing Copilot to measure AI applicability to occupations. We use an LLM-based pipeline to classify the O*NET work activities assisted or performed by AI in each conversation. We find that the most common and successful AI-assisted work activities involve information work—the creation, processing, and communication of information. At the occupation level, we find widespread AI applicability cutting across sectors, as most occupations have information work components. Our methodology also allows us to predict which occupations are more likely to delegate tasks to AI and which are more likely to use AI to assist existing workflows.

1 Introduction

General purpose technologies (1), such as the steam engine and the computer, have historically been strong drivers of economic growth, impacting a broad range of sectors. In the last several years, generative artificial intelligence (AI) has come to the fore as the next general purpose technology (2, 3), capable of improving or speeding up tasks as varied as medical diagnosis (4) and software development (5). These capabilities are reflected in the astounding rate of AI adoption: nearly 40% of Americans report using generative AI, outpacing the early diffusion of the personal computer and the internet (6). Given this widespread adoption and potential for economic impact, a crucial question is *which* work activities AI is most useful for and which occupations perform them, as this will shape where any productivity benefits of AI fall.

We answer these questions by identifying the work activities performed with a mainstream large language model (LLM)-powered generative AI system, Microsoft Bing Copilot (now Microsoft Copilot). A key insight of our analysis is that there are two distinct ways in which a single user–AI conversation is relevant to work activities, corresponding to the two conversing parties. First, the user is seeking assistance with a task they are trying to accomplish; we call this the *user goal*. Analyzing user goals allows us to measure how generative AI is *assisting* different work activities. Second, the AI itself performs a task in the conversation, which we call the *AI action*. Classifying AI actions separately lets us measure which work activities generative AI is *performing*. To illustrate the distinction, if the user is trying to learn how to print a document, the user goal is to operate office equipment, while the AI action is to train others to use equipment. This distinction

*This study was approved by Microsoft IRB #11028. We thank Jennifer Neville, Ashish Sharma, Hancheng Cao, David Holtz, Carolyn Tsao, the Microsoft Research AI Interaction and Learning Group, and the Microsoft Research Computational Social Science Working Group for helpful discussions and feedback, and David Tittsworth, Jonathan McLean, Patrick Bourke, Nick Caurvina, and Bryan Tower for software and data engineering support. Correspondence to: `kitomlinson@microsoft.com, sojaffe@microsoft.com, suri@microsoft.com, counts@microsoft.com`. Results are available at `https://github.com/microsoft/working-with-ai`.

allows to separately identify the types of work activities that workers may delegate to AI, shifting to focus to other tasks, from activities workers may perform in collaboration with AI.

To identify the occupations where AI might be most useful, we draw from a common economic methodology tracing its roots to Autor, Levy, and Murnane (7), which decomposes an occupation into its common tasks and aggregates task-level measurements to occupation-level metrics (3, 8–15). The O*NET database (16) decomposes occupations hierarchically into their work activities, and we use an LLM-based pipeline to map the user goals and AI actions in each conversation to the set of work activities being performed. We measure how successfully different work activities are assisted or performed by AI, using both explicit user feedback and a task completion classifier. To distinguish between broad and narrow AI contributions towards work activities, we also classify the scope of AI capability towards each work activity matched to a conversation. From these classifications, we compute an *AI applicability score* for each occupation. This score captures if there is a non-trivial amount of AI usage that successfully completes activities relevant to an occupation.

Our work connects to research predicting occupational AI impact using various proxies for AI capabilities, including expert forecasts (3, 8), progress on AI benchmarks (9), and AI patents (17). Before the advent of generative AI, similar techniques were used to predict the occupational impacts of robotics, machine learning, and computerization more broadly (7, 10–15, 18). We contribute to this literature by analyzing actual conversations between humans and an LLM, identifying which work activities those humans are using the LLM for and what the associated occupations are. The study most similar to ours is a recent analysis by Handa et al. (19) that classifies Claude conversations using the O*NET taxonomy, but focuses more narrowly on which tasks people use AI for. As a result, they classify each conversation into one O*NET *task*, which is occupation-specific and thereby prevents identification of cross-occupation AI capabilities. Our research goal is to go beyond usage to an estimate of occupational applicability of AI. Thus, we exhaustively classify a conversation into all relevant O*NET *work activities*, which apply across occupations. Furthermore, our measures of conversational success allow us to identify the efficacy of AI across work activities, and our user goal/AI action split allows us to identify how occupations may change in response to AI. More recent work using ChatGPT data adopted our approach of classifying by work activities, but does not use them to measure relevance for occupations (20).

2 Results

2.1 Work activities

We focus our analyses on *intermediate work activities* (IWAs) from the O*NET 29.0 Database (16) (see Section 4 for classification details). We do not know the occupation of a user, but if we observe a work activity being done with AI (regardless of context) then we know it is possible for AI to help with that kind of work. The most common Copilot user goals fall into four broad categories: learning, communicating, teaching/explaining, and writing—with learning and communicating being the most prominent (Figure 1A). The IWAs reflected in AI actions tell a complementary story. Figure 1B shows that the AI plays a service role: some common IWA verbs include *Provide*, *Explain*, *Teach*, *Assist*, and *Respond*. A majority of AI-side activity is devoted to communicating and teaching/explaining information to the user (Figure S2 and table S1 highlight additional differences between user goals and AI actions).

The conversation-level asymmetry between user goals and AI actions is pronounced: these sets of IWAs are disjoint in 40% of conversations, and in 96%, there are more IWAs unique to a side than in common. Furthermore, for every work activity matched to a user goal the AI performs an average of two work activities in service of the user's request (Figure S1).

In O*NET, IWAs are grouped into *generalized work activities* (GWAs); we use GWAs to identify how Copilot usage for work activities compares to the overall frequencies of work activities across the U.S. workforce. Figure 1C shows the activity shares in Copilot aggregated to 37 GWAs and a roughly estimated fraction of total U.S. labor falling under each GWA (described in Appendix B). GWAs focused on learning and communication are most prevalent, echoing the results in Figure 1A,B. The GWAs more prevalent in Copilot usage than in the workforce include *Getting Information*, *Updating and Using Knowledge*, *Communicate*

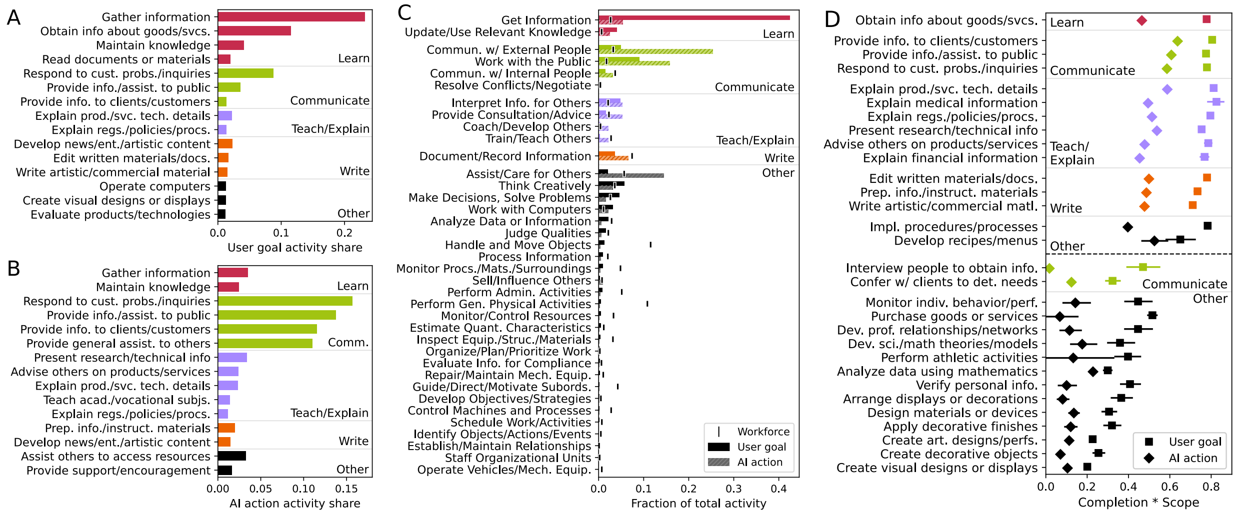

Figure 1: Frequency and success of work activities in Bing Copilot usage. (**A**) The 15 IWAs that occur most frequently as user goals. (**B**) The 15 IWAs that occur most frequently as AI actions. (**C**) The frequencies of GWAs as user goals and AI actions, compared to a rough estimate of GWA workforce frequency (described in Appendix B). (**D**) The top and bottom 15 IWAs (with user goal or AI action activity share at least 0.05%) by the product of their completion rate and the share of conversations where Bing Copilot demonstrated at least a moderate scope of capability. All panels show researcher-generated IWA groupings to highlight patterns.

with External People, and *Interpreting Information for Others*. These align with *information work* (21, 22), concerning the creation, processing, and communication of information, and the related concept of *knowledge work*, which is typically refers to work requiring non-routine and complex problem-solving (23–27). Our data show that the user goals that people use Copilot for, and the activities AI performs in service of those goals, center around information work, and these activities occur disproportionately often relative to the fraction of information work in the workforce.

2.1.1 Metrics of success

Beyond identifying which activities generative AI is being used for, we have three measures of its efficacy: task completion, user feedback, and scope. *Completion* is an LLM-based evaluation of whether the AI completed the user's goal in the conversation validated by real-world *user feedback*, in the form of a thumbs up or down (see Section 4 and Figures S3 to S7). *Scope* is an LLM-based evaluation of the fraction of work in an IWA that Copilot demonstrates the ability to assist or perform, measured via a 6-point ordinal scale (none, minimal, limited, moderate, significant, complete; see Section 4 and Figures S8 and S9). For each IWA, we measure the share of conversations for which the scope is moderate or higher.

Figure 1D shows the top and bottom IWAs ranked by the product of completion rate and scope share. Copilot performs best at activities related to communicating, teaching/explaining, and writing, largely corresponding to the most frequent AI actions in Figure 1B. One the other hand, it performs worse at image generation and data analysis tasks (these patterns are confirmed by thumbs feedback; Figures S3 and S4). Scope in particular is highly correlated with (log) share of user activity ($r = 0.64$; Figure S10). This suggests that users have discovered and are exploiting the informational tasks where AI can have broadest impact. Finally, AI action completion $\times$ scope is consistently lower than on the user goal side, in Figure 1D, indicating that AI can help users with a broader fraction of their work than it can perform directly.

Table 1: **SOC minor groups by AI applicability score.**

Minor Group Title (Abbr.)	Score	Minor Group Title (Abbr.)	Score	Minor Group Title (Abbr.)	Score
Media and Communication Workers*	0.38	Food and Beverage Serving Workers**	0.20	Woodworkers	0.12
Sales Representatives, Services**	0.35	Other Educational and Library Workers*	0.19	Top Executives**	0.11
Information and Record Clerks**	0.33	Material Recording and Distrib. Workers**	0.19	Construction and Extraction Sups.*	0.11
Mathematical Science Occupations	0.32	Prim., Second., and Special Ed. Teachers**	0.18	Assemblers and Fabricators*	0.11
Tour and Travel Guides	0.32	Media and Comms. Equipment Workers	0.18	Printing Workers	0.11
Postsecondary Teachers*	0.31	Advertising, Mktg., PR, and Sales Managers*	0.18	Health Technologists and Technicians**	0.10
Sales Reps., Wholesale and Manufacturing*	0.31	Air Transportation Workers	0.18	Metal Workers and Plastic Workers*	0.10
Communications Equipment Operators	0.30	Lawyers, Judges, and Related Workers*	0.17	Vehicle and Mobile Equipment Mechs.*	0.10
Baggage Porters, Bellhops, and Concierges	0.30	Transportation and Material Moving Sups.*	0.17	Other Maintenance Workers**	0.10
Retail Sales Workers**	0.30	Architects, Surveyors, and Cartographers	0.17	Water Transportation Workers	0.09
Other Sales and Related Workers	0.30	Electrical and Electronic Equipment Mechs.	0.17	Other Production Occupations**	0.08
Computer Occupations**	0.29	Other Healthcare Pracs. and Tech. Occs.	0.16	Bldg. Cleaning and Pest Ctrl. Workers**	0.08
Personal Care and Service Sups.	0.27	Sports Entertainers and Related*	0.16	Firefighting and Prevention Workers	0.07
Entertainment Attendants and Related*	0.27	Other Personal Care and Service Workers*	0.16	Protective Service Sups.	0.07
Religious Workers	0.26	Cooks and Food Preparation Workers**	0.15	Material Moving Workers**	0.07
Social Scientists and Related	0.26	Funeral Service Workers	0.14	Construction Trades Workers**	0.07
Librarians, Curators, and Archivists	0.25	Other Management Occupations**	0.14	Personal Appearance Workers*	0.06
Counselors, Social Workers, and Related**	0.25	Building, Grounds, and Maintenance Sups.	0.14	Agricultural Workers	0.06
Supervisors of Production Workers*	0.25	Other Protective Service Workers*	0.14	Other Construction and Related Workers	0.06
Other Office and Admin. Support Workers**	0.25	Operations Specialties Managers**	0.14	Legal Support Workers	0.06
Office and Administrative Support Sups.*	0.25	Occupational Health and Safety Specialists	0.14	Other Healthcare Support Occupations*	0.06
Financial Clerks**	0.24	Supervisors of Sales Workers*	0.14	Helpers, Construction Trades	0.06
Secretaries and Administrative Assistants**	0.24	Law Enforcement Workers*	0.13	Farming, Fishing, and Forestry Sups.	0.05
Business Operations Specialists**	0.24	Life, Phys., and Social Science Tech'ns	0.13	Occupational and Physical Therapy Assts.	0.05
Animal Care and Service Workers	0.24	Motor Vehicle Operators**	0.13	Textile, Apparel, and Furnishings Workers	0.05
Financial Specialists**	0.23	Healthcare Diagnosing or Treating Pracs.**	0.13	Extraction Workers	0.05
Other Teachers and Instructors*	0.23	Installation, Maintenance, and Repair Sups.*	0.13	Home Health Aides and Nursing Assts.**	0.04
Engineers*	0.22	Rail Transportation Workers	0.12	Grounds Maintenance Workers*	0.04
Physical Scientists	0.21	Other Food Prep. and Serving Workers*	0.12	Plant and System Operators	0.04
Drafters and Eng. and Mapping Tech'ns*	0.21	Food Preparation and Serving Sups.*	0.12	Forest and Conservation Workers	0.03
Life Scientists	0.20	Food Processing Workers*	0.12		
Art and Design Workers*	0.20	Other Transportation Workers	0.12		

Score is the employment-weighted average AI applicability score for each specific occupation in the SOC minor group, averaging the mean of the user goal and AI action scores. Asterisks indicate U.S. employment (none: 0 to 500k, one: 500k to 2M, two: greater than 2M). Minor groups in black have a majority of workers performing information work (classified as described in Appendix F); minor groups in gray have only a minority. Titles have been abbreviated for space (Sups: Supervisors, Pracs: Practitioners).

2.2 AI applicability to occupations

Using O*NET task importance and relevance as weights, we aggregate the IWA-level measures into an *AI applicability score* for each occupation in the Standard Occupational Classification (SOC) (28). The score combines whether Copilot assists or performs an occupation's associated work activities (frequency $\geq 0.05\%$) successfully (completion rate) and in ways that cover a broad share of the work activity (scope $\geq$ moderate). We compute AI applicability score separately for user goals and AI actions, averaging the two when we need a single summary metric. (See Section 4 for a full definition.) Treating all work activities with activity share of at least 0.05% in the same way mitigates potential distributional bias from considering only consumer Copilot data. The use of a frequency threshold means relative comparisons are more meaningful than absolute score values (see Section 4 and Figures S19 and S20).

The use of AI for information work activities observed in Figure 1 translates directly to the occupations where AI is most applicable. Figure 2 shows the 25 occupations with the highest AI applicability score and the work activities that contribute the most to those occupations' scores. The top IWAs involve the creation (e.g., *Edit written materials, Write commercial material*), gathering (e.g., *Maintain knowledge*), and dissemination (e.g., *Respond to customer inquiries, Provide information to customers*) of information. In fact, almost all of the top-contributing IWAs fall into one or more of these categories. These IWAs contribute to the AI applicability scores of a number of occupations involving information creation and processing, such as Interpreters and Translators, Writers, Editors, and Data Scientists. Occupations involving the communication and dissemination of information also stand out with high AI applicability, such as Sales Representatives, Customer Service Representatives, and Concierges.

To get a broader view of all occupations in the economy, Table 1 reports the AI applicability of all SOC minor groups, highlighting occupational groups containing a majority of workers in occupations that consist primarily of information work (Appendix F describes how we classify occupations as information work). The

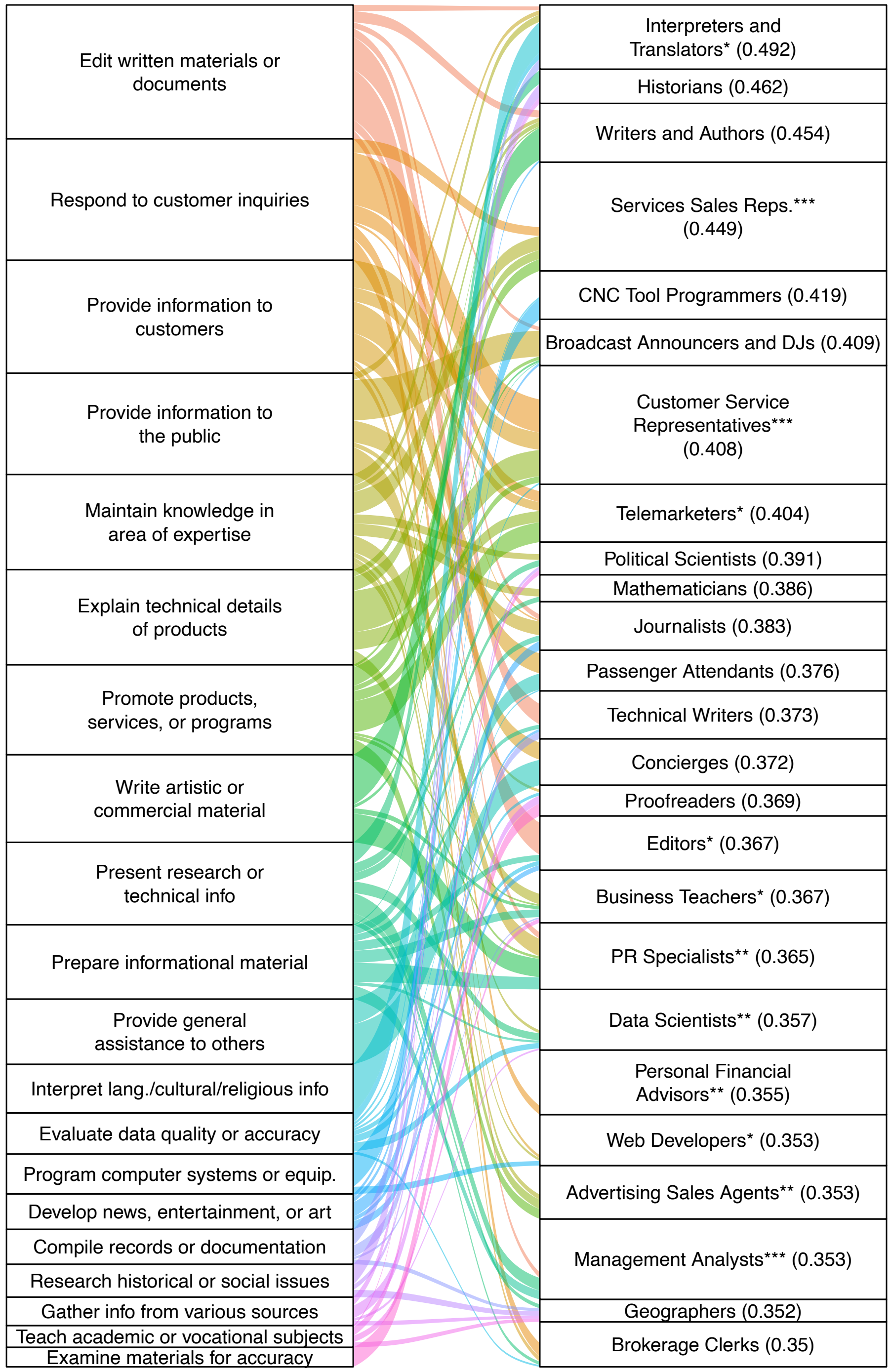

Figure 2: Top occupations by AI applicability score and their contributing IWAs. The 25 SOC occupations with the highest AI applicability scores (right) along with the 20 IWAs that provide the greatest contributions to those scores (left). Asterisks indicate 2023 U.S. occupational employment (none: up to 50k, one: 50k to 100k, two: 100k to 500k, three: more than 500k); the heights of each stratum indicate the same. Occupations' applicability scores are in parentheses, and occupations are sorted by their score in decreasing order. Portions of the occupational strata not connected to an IWA by a colored flow represent IWAs not present in the Figure that still contribute to the occupation's applicability score. Both occupation and IWA titles have been shortened for space.

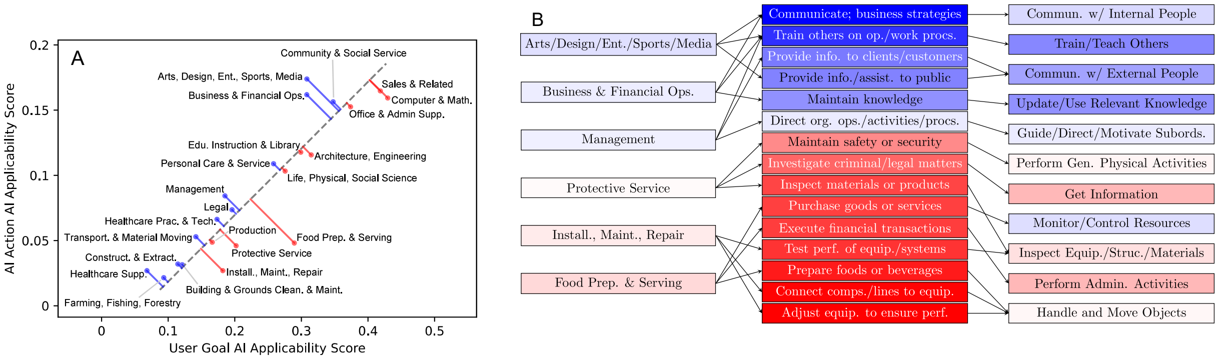

Figure 3: Differences between occupations' user-goal and AI-action applicability and the contributing work activities. (**A**) Scatterplot of each major occupation group's (employment-weighted) average AI action applicability score and average user goal applicability score. The dashed gray line shows the first principal component, capturing overall applicability. The blue and red lines show the extent to which a major group has higher applicability from one metric versus the other (based on the second principal component, which captures the extent to which, compared to the overall relationship between user score and AI score, the score skews more toward the red user-goal or blue AI-action side). (**B**) A tripartite graph showing six prominent occupation groups that are skewed toward user goal or AI action applicability (left), the three IWAs contributing the most to each group's skew (middle) ordered by the extent to which those IWAs skew more to the AI action side, and the corresponding GWAs. As in (A), blue indicates a larger skew toward AI action applicability, while red indicates a larger skew toward user goal applicability. White is neutral. See Section 4 for further details.

top groups are mostly information workers, with Media and Communication, Information and Record Clerks, and Sales Representatives of Services at the very top. The groups with the lowest scores include occupations that require physically working with people, operating machinery, and other manual labor. However, we find that most occupations have at least some AI applicability, reflecting that most work has some information processing component (21).

At a high level, these observations of AI applicability based on real-world usage align with predictions of AI capabilities. Eloundou et al. (3) used six annotators to predict the share of tasks within each occupation for which access to an LLM alone would lead to time savings. The employment-weighted occupation-level correlation between their predictions and our AI applicability score is $r = 0.73$ (Figure S12). But our data do not fully confirm expectations: for instance, we find only a weak relationship between AI applicability score and wage (employment-weighted $r = 0.13$; Figure S13A), while previous work found strong positive correlations (3, 9) (some of this may be due to a lack of employment-weighting in prior work; see Appendix G). Similarly, we find a broad range of AI applicability across educational requirements (Figure S13B). Beyond wage and education, AI may have differential effects across demographic groups to the extent that some demographics are more or less likely to be employed in AI-applicable occupations (Figure S18).

2.2.1 User goals and AI actions

Our approach of separating user goals from AI actions allows us to not only identify where AI is most useful, but also separate out the different ways AI may change the way occupations do their work. There are at least two different ways occupations could change as a result of AI: (1) delegating some tasks to AI, freeing up workers to focus on different parts of the job, corresponding to high AI action applicability, and (2) performing the same tasks but in collaboration with AI, corresponding to high user goal applicability score. Figure 3A plots the average AI-action and user-goal applicability scores for each major SOC group, highlighting occupational groups that lean towards either side of this dichotomy. The lengths of the red (resp. blue) lines indicate the extent to which the occupations in that group are more associated with user goals (resp. AI actions); these are derived from the second principal component of the user goal and AI action applicability scores (see Section 4). To see which activities are driving these differences, Figure 3B takes three of the groups with the largest difference in each direction (left column), maps them to the IWAs that contribute most to that difference (middle column), and then shows which GWAs those IWAs are part of (right column). Figure 3 indicates that media and financial operations occupations are more likely to delegate tasks to AI, such as those involving communication and training. On the other hand, to the extent that AI is used for more physical occupations like those in the food preparation and serving category, it is likely to be in a more assistive role.

3 Discussion

This work provides an early signal indicating where and how AI is likely to change the way people work. Our data shows that people use AI to do a variety of work activities related to the creation, analysis, and processing of information, i.e., information work. Most occupations have at least some information work component, reflected in our finding that AI is somewhat relevant to most occupations. This helps to explain the rapid rate of adoption of AI in the workplace (6). Our analysis also highlights the different ways AI may be used in different roles. Occupations with high AI applicability to user goals, such as computer and mathematical occupations, may see more active AI usage by workers as they use AI to improve existing workflows. Meanwhile, occupations with high AI action applicability, like those in business and financial operations, may delegate tasks to AI and focus more on the parts of their jobs that require a human touch.

Understanding which occupations have tasks within the current frontier of AI capability is important because these jobs are most likely to see productivity boosts or a shift in their core tasks. In addition, understanding the current frontier of AI applicability allows system designers to investigate where AI is currently falling short and see if it is due to an addressable technical shortcoming (such as low satisfaction for data analysis), a mismatch in modality (such as moving objects), or a societal factor (such as laws or regulations). Looking ahead, as AI systems gain functionality and adoption continues to evolve, we can use our AI applicability scores to see how the AI frontier shifts over time.

Our results show that AI is most applicable to information work occupations and often plays the role of an advisor or teacher, providing expertise to the user. In the same way that word processors turned typing into a widespread skill rather than a core task for typists and secretaries, AI too may democratize skills around information work. To the extent that people can successfully apply AI expertise, shrinking the performance gap between low- and high-skilled workers (5, 29–31), this provides some empirical backing to Autor's theoretical argument (32) that AI could reduce income inequality: if AI can broaden access to expertise, and people have sufficient foundational knowledge to apply and evaluate AI assistance, then AI could provide more workers access to previously niche expert work, potentially shrinking economic inequality.

Our user goal versus AI action distinction relates to a key question in the literature and public discourse around AI: to what extent is AI automating versus augmenting work activities? It is tempting to conclude that occupations that have high AI action applicability score will be automated and thus experience job or wage loss, and that occupations with high user goal applicability score will be augmented and raise wages. This would be a mistake, as downstream consequences of new technologies are very hard to predict and often counterintuitive (33). Take the example of ATMs, which automated a core task of bank tellers, but led to an *increase* in the number of bank teller jobs as banks opened more branches at lower costs and tellers focused on more valuable relationship-building rather than processing deposits and withdrawals (34). Our measures of AI applicability indicate where AI is likely to change work, which is a part of the automation vs. augmentation question, but AI's effects on employment and wages will depend on hard-to-predict business decisions.

There are some natural limitations to the conclusions that can be drawn from our data. Our task completion and scope metrics are measures of AI's utility towards an IWA, but they do no capture the productivity impact on people doing that IWA. Another gap is the difference between the way work activities are performed in occupations compared to in our data (for instance, *Provide general assistance* means something different for a passenger attendant and for Copilot). Our data also represents only one slice of the AI market: there are many other AI platforms, including more task- or occupation-specific LLMs, which are not represented in our data. While decomposing an occupation into its work activities is standard practice in the literature, it does not provide a complete representation of every occupation: the connecting glue between tasks also contributes to the value of work. Finally, our use of O*NET means our results are shaped by its U.S.-centric view, may lag behind current workplace activities, and do not capture valuable tasks performed outside of occupations (e.g., work in the home or volunteering). Modernizing our understanding of workplace activities will be crucial as generative AI continues to change how work is done.

This work gives rise to a number of future research questions of high importance to society. We measured how AI capabilities overlap with work activities, but it remains to be seen how different occupations refactor their work responsibilities in response to AI's rapid progress. In addition, entirely new occupations may emerge, performing new types of work activities (35, 36). This is not a new phenomenon: the *majority* of employment today is in occupations that arose in the last 100 years as a result of new technologies (37). Which new jobs emerge, and how old ones are reconstituted, is a crucial future research direction.

Information work has become central to the global economy, embedded in the majority of occupations. Technological advances of the last several decades, such as the computer, the internet, and the search engine, have provided valuable tools for storing, processing, and finding information more efficiently. Our results highlight how LLMs can contribute to broader parts of the information life cycle—including creation, interpretation, and communication, in more flexible ways than earlier technologies. As generative AI continues to progress, it will be important to map the frontier of AI functionality to understand what additional capabilities AI is providing people and how that new functionality can be applied in the workplace.

4 Materials and methods

4.1 Bing Copilot data

We analyze anonymized and privacy-scrubbed U.S. conversation data from Microsoft Bing Copilot (henceforth, Copilot) gathered from January 1, 2024 to September 30, 2024. We focus only on conversations in the United States to align with occupation and work activity information from O*NET. Our main dataset is a uniform sample of approximately 100k conversations, providing a representative view of what tasks users perform with a mainstream, publicly available, free-to-use generative AI chatbot. We use a supporting dataset of 100k conversations uniformly sampled from those that received at least one thumbs up or thumbs down reaction from the user. This dataset allows us to investigate what activities are performed more or less successfully, as measured by explicit user feedback. Our use of Copilot data was reviewed and approved by the Microsoft IRB (under ID # 11028). While we cannot release conversational data due to legal and privacy obligations, all of our aggregated IWA- and occupation-level metrics have been made public on GitHub (`https://github.com/microsoft/working-with-ai`).

4.2 O*NET and Bureau of Labor Statistics data

We use O*NET's (16) hierarchical decomposition of occupations into their tasks and work activities. Our analysis focuses on *intermediate work activities* (IWAs), which map to multiple occupations through tasks. We combine O*NET with data on wages and employment from the 2023 Occupational Employment and Wage Statistics data published by the U.S. Bureau of Labor Statistics (BLS) (38). We use task frequency and relevance data from O*NET to get a rough frequency of how often each IWA is done by the U.S. labor force (details in Appendix B). We also use wage, education, and demographic data from the Current Population Survey (CPS) data from 2024 (39), which is gathered by the U.S. Census Bureau for the BLS. All of our occupational analysis is performed on occupations from the 2018 Standard Occupational Classification (SOC) (28). Appendix A describes how we map the O*NET and CPS occupational taxonomies to SOC codes and Appendix A.1 describes which SOC codes we drop due to missing data.

4.3 Work activity classification

For each conversation in our datasets, we use an LLM classification pipeline to identify *all* IWAs that match the user goal and the AI action. In most conversations, there are multiple IWA matches on both sides (on average, about 3 user goal IWAs and 6 AI actions IWAs; see Figure S1). If the user goal or AI action is not related to any work activity, then it should be matched with zero IWAs. All prompts, pipeline architecture, model details, and human validation metrics are provided in Appendix C.

4.3.1 Activity share and coverage

Since each conversation can be assigned multiple IWAs, instead of counting raw IWA occurrences in our conversation data, we focus on each IWA's *activity share*, where we allocate an equal fraction of each conversation to its labeled IWAs, separately on the user and AI sides. We consider the work activities that appear at non-trivial frequencies in Copilot chat data to be "covered." We define non-trivial usage using a threshold of 0.05% activity share and use this as a signal that AI can potentially assist or perform that IWA. We chose the threshold 0.05% to minimize the number of occupations with no or all IWAs covered, thereby maximizing the usefulness of the measure for relative comparisons between occupations (Figure S19A). The ordering of occupations induced by our AI applicability score is robust to the chosen coverage threshold (Figure S19B). We define an occupation's *coverage* to be the weighted fraction of its work activities that are covered (using the occupation–IWA weights defined in Section 4.4.1; see Figures S21 and S22).

4.3.2 Completion and Scope

We use an conversation-level LLM completion classification (details and prompts in Appendix D), to measure the share of conversations labeled with each IWA where the AI completed the user's task in the conversation. Completion is highly correlated with direct user feedback (weighted $r > 0.75$; Figure S5), and has the advantage of being available for all conversations, so it is not subject to the selection bias of thumbs feedback.

For each relevant IWA in a conversation, we also perform an LLM classification of the fraction of work in the IWA that Copilot demonstrates the ability to assist or perform, which we call the *scope*, measured on a six-point ordinal scale: none, minimal, limited, moderate, significant, complete. The goal of scope is to distinguish between cases where Copilot assists with a large or small fraction of the work in an IWA. Scope classification and validation are detailed in Appendix C.

4.4 Occupational AI applicability score

4.4.1 Aggregating from IWAs to occupations

We compute a weight for every task using the importance and relevance scores in O*NET, which allows us to capture which work activities are more or less important for each occupation. For each task k in SOC occupation i, we say $\text{weight}_{ik} = 2^{\text{importance}_{ik}} \cdot \text{relevance}_{ik}$. We sum weights for tasks mapping to the same IWA. When a task maps to multiple IWAs, its weight contributes to them equally. If an occupation has no ratings for any of its tasks, we assign them all weight $w_{ik} = 1$. If an occupation has ratings for only some of its tasks, we ignore the tasks with missing ratings. Dividing IWA weights by the total weight for an occupation then gives us a proxy measure for how much of a job consists of each of its IWAs. We use w_{ij} to denote this normalized weight of IWA j in job i.

In addition, we also compute a second IWA weight w_{ij}^{nonphys} to account for the fact that some IWAs like *Provide general assistance* map to both physical and nonphysical tasks, but applicability to the IWA does not imply AI action applicability to physical tasks. To compute w_{ij}^{nonphys}, we use GPT-5 to label every O*NET task according to whether it requires touching or moving people or objects, validated by two human annotators (prompt and validation details in Appendix E). When summing task weights w_{ik} to get w_{ij}^{nonphys}, we only include those tasks labeled as nonphysical, but normalize by the weight of all tasks; this makes w_{ij}^{nonphys} as estimate of the fraction of an job i's work that is both nonphysical and described by IWA j.

4.4.2 AI applicability score

We aggregate IWA coverage, completion, and scope into an AI applicability score for each occupation. For the user-goal side,

$$a_i^{\text{user}} = \sum_{j \in \text{IWAs}(i)} \mathbf{1}[f_j^{\text{user}} \geq 0.0005] c_j^{\text{user}} s_j^{\text{user}} w_{ij}, \tag{1}$$

where IWAs(i) is the set of IWAs performed by occupation i, $f_j^{\text{user}}$is the user goal activity share of j, c_j^{user} is the task completion rate of conversations with IWA j as a user goal, and s_j^{user} is the fraction of conversations with user goal j in which the scope classification is moderate or higher. We define a_i^{AI} similarly for AI actions but using AI action IWA measures and, for a_i^{AI} only, using w_{ij}^{nonphys} instead of w_{ij}. We report $a_i = (a_i^{\text{user}} + a_i^{\text{AI}})/2$ unless otherwise specified.

Our approach contrasts with other research that uses a measurement (19) or prediction (3) of the fraction of occupations or of the workforce that have at least x% of their tasks impacted by AI. Such measurements cannot be made reliably from usage data alone, as the selected threshold for usage has a significant impact on the resulting numbers, whose apparent straightforwardness belies this issue. Figure S20 shows that by picking different usage thresholds, we can conclude that either $\sim 0\%$ or $\sim 100\%$ of the workforce has 50% of its importance-weighted tasks represented in our data, depending on whether we require 1% of chat activity for a task to be covered or only .01% of activity. As such, we believe it is much more meaningful to make relative statements about different kinds of occupations, which is what our AI applicability score is designed to do.

4.4.3 User goal vs. AI action applicability

We use principal component analysis (PCA) to measure the extent to which work activities or occupations are more successfully represented in user goals or AI actions. We either perform PCA on occupational AI applicability scores (as in Figure 3A) or on IWA applicability scores (as in Figure 3B), where an IWA j's (user goal) applicability score is $\mathbf{1}[f_j^{\text{user}} \geq 0.0005] c_j^{\text{user}} s_j^{\text{user}}$ (i.e., its contribution to Equation (1), before occupation-specific weighting). As the user goal and AI scores are strongly correlated, the first principal component captures overall applicability, while the second perpendicular component measures skew towards AI actions or user goals. To identify which IWAs contribute most to an occupation group's skew in Figure 3B, we take the second component magnitudes in IWA-level PCA, multiply them by IWA weights w_{ij} for each occupation, and then take an employment-weighted average to get a score for each occupational group × IWA. IWA-level scores are also averaged to get GWA-level scores.